Untitled

2026-11-03

```
zapisw <- c()
zapist <- c()
for (i in delta){
  countw <- 0
  countt <- 0
  for (j in 1:M){
    x <- rnorm(n)
    y <- rnorm(m, i)

    w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
    t <- t.test(x,y,alternative="less",var.equal = TRUE)

    if (w$p.value < alpha){countw <- countw + 1}
    if (t$p.value < alpha){countt <- countt + 1}
  }
  zapist <- c(zapist, countt / M)
  zapisw <- c(zapisw, countw / M)
}
```

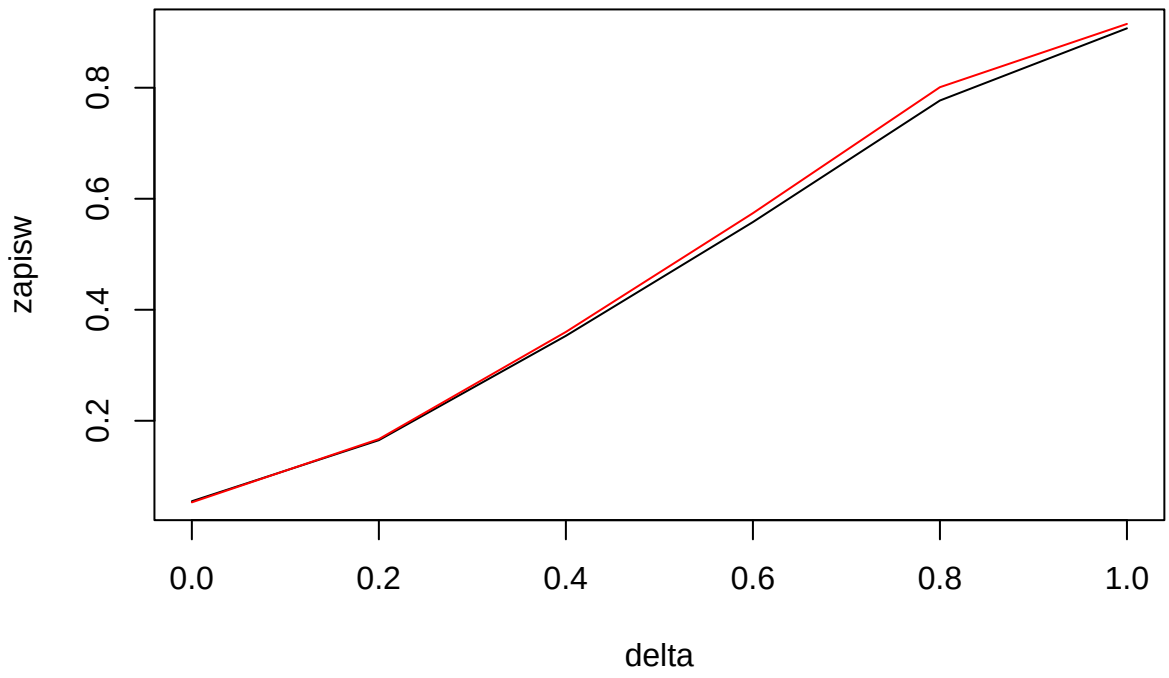
```
cat("delta Moc testu Wilcoxona Moc testu t-Studenta Lepszy test\n")
```

```
## delta Moc testu Wilcoxona Moc testu t-Studenta Lepszy test
```

```
for (i in 1:6){
  lepszy <- ""
  if(zapist[i] < zapisw[i]){
    lepszy <- "W"
  }
  else{lepszy <- "T"}
  cat(zapist[i]," ",zapisw[i]," ",lepszy ,"\n")
}
```

```
## 0.053   0.055   W
## 0.167   0.165   T
## 0.36   0.353   T
## 0.574   0.558   T
## 0.801   0.777   T
## 0.915   0.907   T
```

```
plot(delta, zapisw, type = "l")
lines(delta, zapist, col="red")
```




```
P95 <-function(){rbinom(m,1,0.95)}
P90 <-function(){rbinom(m,1,0.90)}
N05 <-function(){rnorm(m,0.5,1)}
N8 <-function(){rnorm(m,8,1)}
N12 <-function(){rnorm(m,12,1)}
N20 <-function(){rnorm(m,20,1)}
```

```
zapisw2 <- c()
zapist2 <- c()

countw <- 0
countt <- 0
for (j in 1:M){
  x <- rnorm(n)
  y <- rnorm(m, 0.5)

  w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
  t <- t.test(x,y,alternative="less",var.equal = TRUE)

  if (w$p.value < alpha){countw <- countw + 1}
  if (t$p.value < alpha){countt <- countt + 1}
}
zapist2 <- c(zapist2, countt / M)
zapisw2 <- c(zapisw2, countw / M)
```

```
countw <- 0
countt <- 0
```

```
for (j in 1:M){
  x <- rnorm(n)
  temp <- P95()
  y <- temp*N05() + (1-temp)*N8()

  w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
  t <- t.test(x,y,alternative="less",var.equal = TRUE)

  if (w$p.value < alpha){countw <- countw + 1}
  if (t$p.value < alpha){countt <- countt + 1}
}
zapist2 <- c(zapist2, countt / M)
zapisw2 <- c(zapisw2, countw / M)
```

```
countw <- 0
countt <- 0
for (j in 1:M){
  x <- rnorm(n)
  temp <- P90()
  y <- temp*N05() + (1-temp)*N8()

  w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
  t <- t.test(x,y,alternative="less",var.equal = TRUE)

  if (w$p.value < alpha){countw <- countw + 1}
  if (t$p.value < alpha){countt <- countt + 1}
}
zapist2 <- c(zapist2, countt / M)
zapisw2 <- c(zapisw2, countw / M)
```

```
countw <- 0
countt <- 0
for (j in 1:M){
  x <- rnorm(n)
  temp <- P95()
  y <- temp*N05() + (1-temp)*N12()

  w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
  t <- t.test(x,y,alternative="less",var.equal = TRUE)

  if (w$p.value < alpha){countw <- countw + 1}
  if (t$p.value < alpha){countt <- countt + 1}
}
zapist2 <- c(zapist2, countt / M)
zapisw2 <- c(zapisw2, countw / M)
```

```
countw <- 0
countt <- 0
for (j in 1:M){
  x <- rnorm(n)
  temp <- P90()
  y <- temp*N05() + (1-temp)*N12()

  w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
```

```
  t <- t.test(x,y,alternative="less",var.equal = TRUE)

  if (w$p.value < alpha){countw <- countw + 1}
  if (t$p.value < alpha){countt <- countt + 1}
}
zapist2 <- c(zapist2, countt / M)
zapisw2 <- c(zapisw2, countw / M)
```

```
countw <- 0
countt <- 0
for (j in 1:M){
  x <- rnorm(n)
  temp <- P95()
  y <- temp*N05() + (1-temp)*N20()

  w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
  t <- t.test(x,y,alternative="less",var.equal = TRUE)

  if (w$p.value < alpha){countw <- countw + 1}
  if (t$p.value < alpha){countt <- countt + 1}
}
zapist2 <- c(zapist2, countt / M)
zapisw2 <- c(zapisw2, countw / M)
```

```
cat("delta Moc testu Wilcoxona Moc testu t-Studenta Lepszy test\n")
```

```
## delta Moc testu Wilcoxona Moc testu t-Studenta Lepszy test
```

```
for (i in 1:6){
  lepszy <- ""
  if(zapist2[i] < zapisw2[i]){
    lepszy <- "W"
  }
  else{lepszy <- "T"}
  cat(zapist2[i]," ",zapisw2[i]," ",lepszy, "\n")
}
```

```
## 0.48   0.463   T
## 0.589   0.531   T
## 0.729   0.632   T
## 0.567   0.535   T
## 0.713   0.644   T
## 0.451   0.518   W
```

Nie jest spełnione założenie o normalności wiec dla duzych sie rozjedza. Dla małych sa male srednie wiec nie wykrywa

```
zapisw <- c()
zapist <- c()
for (i in delta){
  countw <- 0
  countt <- 0
  for (j in 1:M){
    x <- rt(n, df=3)
    y <- rt(m, df=3) + i
```

```
    w <- wilcox.test(x,y,alternative="less",exact=FALSE,correct=FALSE)
    t <- t.test(x,y,alternative="less",var.equal = TRUE)

    if (w$p.value < alpha){countw <- countw + 1}
    if (t$p.value < alpha){countt <- countt + 1}
  }
  zapist <- c(zapist, countt / M)
  zapisw <- c(zapisw, countw / M)
}
```

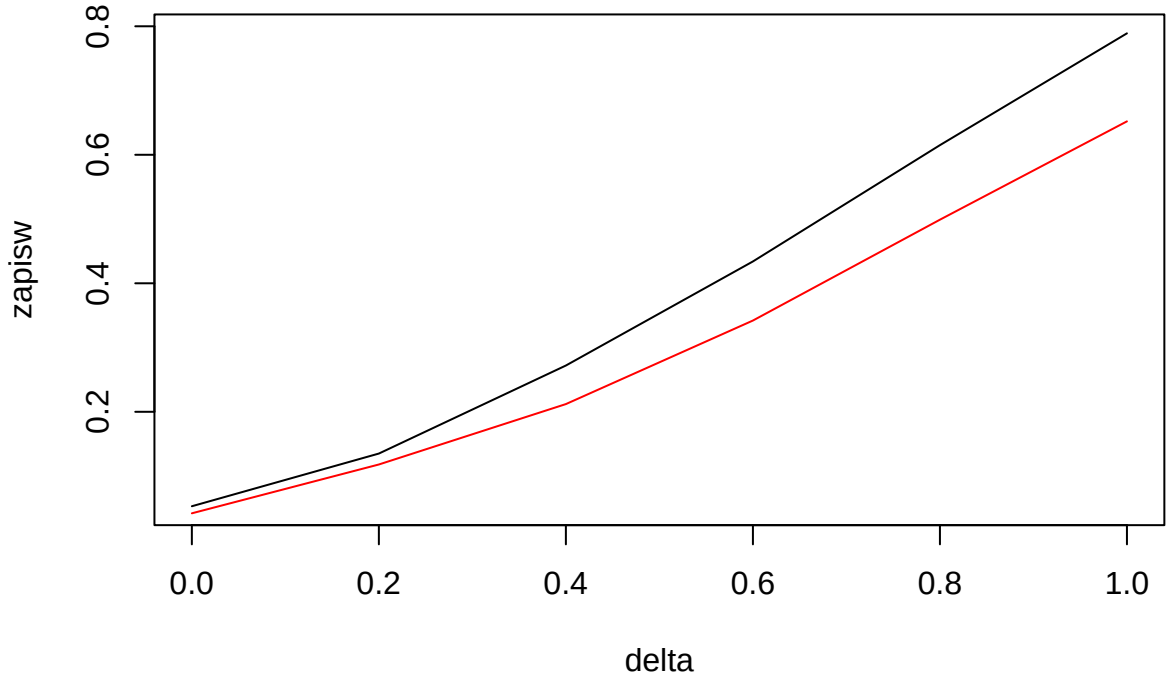
```
cat("delta Moc testu Wilcoxona Moc testu t-Studenta Lepszy test\n")
```

```
## delta Moc testu Wilcoxona Moc testu t-Studenta Lepszy test
```

```
for (i in 1:6){
    lepszy <- ""
  if(zapist[i] < zapisw2[i]){
    lepszy <- "W"
  }
  else{lepszy <- "T"}
  cat(zapist[i]," ",zapisw[i]," ",lepszy, "\n")
}
```

```
## 0.042   0.053   W
## 0.118   0.135   W
## 0.212   0.272   W
## 0.342   0.434   W
## 0.499   0.615   W
## 0.652   0.789   T
```

```
plot(delta, zapisw, type = "l")
lines(delta, zapist, col="red")
```



Test Wilcoxona dziala na rangach więc zamineia obserwacje odstające na porostu rangi. Test wtedy nie widzi tych odleglosci.